

Feedback-Dressed Dynamic Topology of Inverter-Dominated Power Systems: Hidden Cycles, Reinforcement Reversals, and Controller-Aware Screening

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Abstract—Conventional grid-strength metrics characterize the physical electrical network, whereas inverter-rich stability is set by a closed-loop network that also contains device and controller dynamics. This paper develops a feedback-dressed interpretation of that discrepancy. Controller self-energy $\Sigma(s)$ is composed with the frequency-dependent network resolvent $G_r(s)$ to form $K_H(s) = G_r(s)\Sigma(s)$; the controller-side return ratio $R_c(s)$ has the same nonzero loop spectrum, and the full characteristic determinant factors exactly into local network dynamics, local controller dynamics, and an interaction determinant. Schur condensation exposes frequency-dependent effective edges and cycles that need not exist in the retained physical graph. On a transparent IEEE 39-bus proof-of-concept, reinforcing branch 19–20 by $1.8\times$ improves algebraic connectivity by 0.694%, reduces a bus-34 network-impedance proxy by 4.60%, and reduces AC losses by 0.227%, yet collapses the hidden feedback margin by 91.3% and reduces controller-family damping from 2.89% to 0.186%. A passive reduced cycle gain of 0.00596 becomes 1.013 after controller dressing and falls to 0.0787 after controller retuning while the physical reinforcement is retained. The mechanism survives loading sweeps, five retained-port choices, continuous controller-tuning sweeps, action-strength variation, nonlinear disturbance spectra, and parameter perturbations. Across 223 base-stable parameter draws, damping deteriorates in every draw and 31.4% cross instability after reinforcement. Most importantly, an all-action nonlinear cross-check applies identical broadband stochastic forcing to every one of the 46 reinforced branches: the linear hidden margin predicts nonlinear controller-action RMS with Spearman $\rho = -0.851$, while the full-loop spectral radius preserves the nonlinear ordering with $\rho = 0.994$. Four-thousand-bootstrap intervals for physical λ_2 , Fiedler-only projection, and raw Σ mixing overlap zero, whereas feedback-dressed cross-mode coupling, full-loop spectrum, and hidden margin do not. These results support controller-aware planning based on physical strength plus dynamic feedback topology rather than either view alone, while independent ANDES/EMT validation remains a required next test.

Index Terms—inverter-based resources, grid-following inverter, small-signal stability, graph signal processing, Fiedler vector, Schur complement, self-energy, hidden dynamic graph, feedback cycle, transmission planning.

I. INTRODUCTION

LOW-INERTIA and inverter-dominated grids are often screened with quantities derived from the physical network: short-circuit strength, impedance, loading, algebraic

connectivity, Laplacian eigenvectors, or reduced L – M – D models. These quantities are useful because they summarize the network seen by power-flow or electromechanical coordinates. They are not, however, a complete representation of the feedback paths created by phase-locked loops (PLLs), current controllers, outer loops, exciters, governors, stabilizers, and other dynamic devices.

Impedance-based stability criteria already establish that grid and converter dynamics must be analyzed jointly [2], and interaction-admittance models extend that view to multi-inverter networks [3]. PLL–grid coupling is also known to change small-signal stability and can require controller-side remedies [4]. Separately, Kron reduction formalizes how eliminating electrical nodes creates effective edges in reduced networks [5], and graph signal processing (GSP) provides a spectral language for signals and operators on graph modes [7]. The question pursued here is narrower: *how should a physical topology action be mapped to the closed-loop dynamic feedback that actually determines an inverter-rich stability margin?*

The key observation is that controller dynamics are not stability-relevant merely because they exist. Their effect must be propagated through the dynamic susceptibility of the network and returned through the feedback loop. We therefore distinguish four objects:

$$L_P \longrightarrow \Sigma(s) \longrightarrow K_H(s) = G_r(s)\Sigma(s) \longrightarrow R_c(s), \quad (1)$$

where L_P describes the physical graph, $\Sigma(s)$ is the controller self-energy on retained network states, $G_r(s)$ is the network resolvent, $K_H(s)$ is the feedback-dressed network operator, and $R_c(s)$ is the controller-side return ratio. The proposed *hidden margin* is based on the distance of $I - R_c(j\omega)$ from singularity.

The algebra used below—Schur complements, determinant factorizations, Woodbury updates, and spectral graph projections—is standard. The proposed contribution is their specialization into a single power-system workflow from *physical action* to *hidden feedback topology* to *dynamic consequence*. The main contributions are:

- an exact determinant factorization that separates local network and controller dynamics from interaction-induced modes;

- a feedback-dressed effective graph whose hidden edges and cycles are generated by Schur condensation and controller/network propagation;
- a low-rank topology-to-feedback map for transmission actions, together with a controller-side hidden feedback margin;
- a GSP interpretation showing why a Fiedler-only or raw- Σ view is insufficient, while feedback-dressed cross-mode coupling is more informative;
- an exhaustive IEEE 39-bus proof-of-concept in which electrically beneficial reinforcements are ranked before nonlinear simulation and the frozen ranking predicts nonlinear controller stress;
- falsification-oriented robustness tests across loading, controller tuning, retained-port selection, action magnitude, nonlinear disturbance spectrum, and parameter uncertainty, with bootstrap confidence intervals for the predictor hierarchy.

Recent work reinforces the need to distinguish these layers. Full-order holistic stability models retain coupled network and control dynamics in large inverter-intensive systems [10]; frequency-dependent system-strength assessment explicitly shows that “strength” is not a single static scalar [11]; whole-system admittance identification provides a practical closed-loop route to oscillation and root-cause analysis [12]; and a 2026 graph-theoretic stability framework shows how topology, operating point, and inverter dynamics can be combined for a different class of lossless GFM models [13]. These results delimit the present claim: the novelty pursued here is not frequency dependence, Schur reduction, or graph analysis individually, but an action-resolved map from a physical reinforcement to a feedback-dressed effective topology, its hidden cycle closure, and a falsifiable planning screen.

The paper intentionally does not claim independent DAE/EMT validation. MATPOWER [8] is used for the AC operating point and physical-network metrics; ANDES [9] is identified as the next independent validation environment.

II. PHYSICAL AND DYNAMIC NETWORK MODELS

A. Physical AC graph and Fiedler coordinates

Let a solved AC operating point be denoted by $x^* = (\theta^*, V^*, \dots)$. A local physical coupling matrix can be formed from the angle Jacobian or, in the lossless approximation,

$$L_P(x^*) = BW(x^*)B^\top, \quad W_{ee} = b_e V_i^* V_j^* \cos(\theta_i^* - \theta_j^*), \quad (2)$$

where B is the branch incidence matrix. Its ordered eigenpairs satisfy

$$L_P U = U \Lambda, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots, \quad (3)$$

with λ_2 the algebraic connectivity and u_2 the Fiedler vector [6]. The graph Fourier transform of a bus signal y is $\hat{y} = U^H y$ [7]. These coordinates provide a natural baseline for asking whether controller dynamics preserve or mix the physical graph modes.

B. Linearized DAE and controller self-energy

Consider a linearized dynamic model after algebraic elimination,

$$\dot{x} = Ax, \quad A = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix}, \quad (4)$$

where r denotes retained network/interface states and c controller/device states. Define

$$T_r(s) = sI - A_{rr}, \quad T_c(s) = sI - A_{cc}, \quad (5)$$

and the network-side controller self-energy

$$\Sigma_r(s) = A_{rc} T_c(s)^{-1} A_{cr}. \quad (6)$$

The exact retained operator is

$$T_{\text{eff},r}(s) = T_r(s) - \Sigma_r(s). \quad (7)$$

For electromechanical retained coordinates, (7) contains the familiar second-order form $s^2 M + s \Sigma_D(s) + L$ as a special representation, but the state-space form is used here because it does not require controller dynamics to be collapsed into a scalar damping coefficient.

C. Feedback-dressed network operator and hidden margin

Let

$$G_r(s) = T_r(s)^{-1}, \quad G_c(s) = T_c(s)^{-1}. \quad (8)$$

We define the feedback-dressed network operator

$$K_H(s) = G_r(s) \Sigma_r(s) = G_r A_{rc} G_c A_{cr} \quad (9)$$

and the controller-side return ratio

$$R_c(s) = G_c(s) A_{cr} G_r(s) A_{rc}. \quad (10)$$

The proposed hidden feedback margin is

$$d_H = \min_{\omega \in \Omega} \sigma_{\min}[I - R_c(j\omega)]. \quad (11)$$

Unlike a physical strength index, d_H is a closed-loop quantity: it becomes small when a network-mediated controller return approaches singular feedback closure.

III. ALGEBRAIC STRUCTURE OF HIDDEN DYNAMIC TOPOLOGY

A. General interaction determinant

Proposition 1 (Interaction determinant): For frequencies where $T_r(s)$ and $T_c(s)$ are nonsingular,

$$\det(sI - A) = \det T_r(s) \det T_c(s) \det[I - R_c(s)]. \quad (12)$$

Equivalently,

$$\det(sI - A) = \det T_c(s) \det T_r(s) \det[I - K_H(s)]. \quad (13)$$

Proof: Apply the Schur determinant identity to $sI - A$ using either diagonal block. The second equality follows from Sylvester’s determinant theorem $\det(I - XY) = \det(I - YX)$. ■

Equation (12) separates local poles of the network and controller blocks from interaction-induced zeros of $\det(I - R_c)$. This is the precise sense in which the proposed hidden

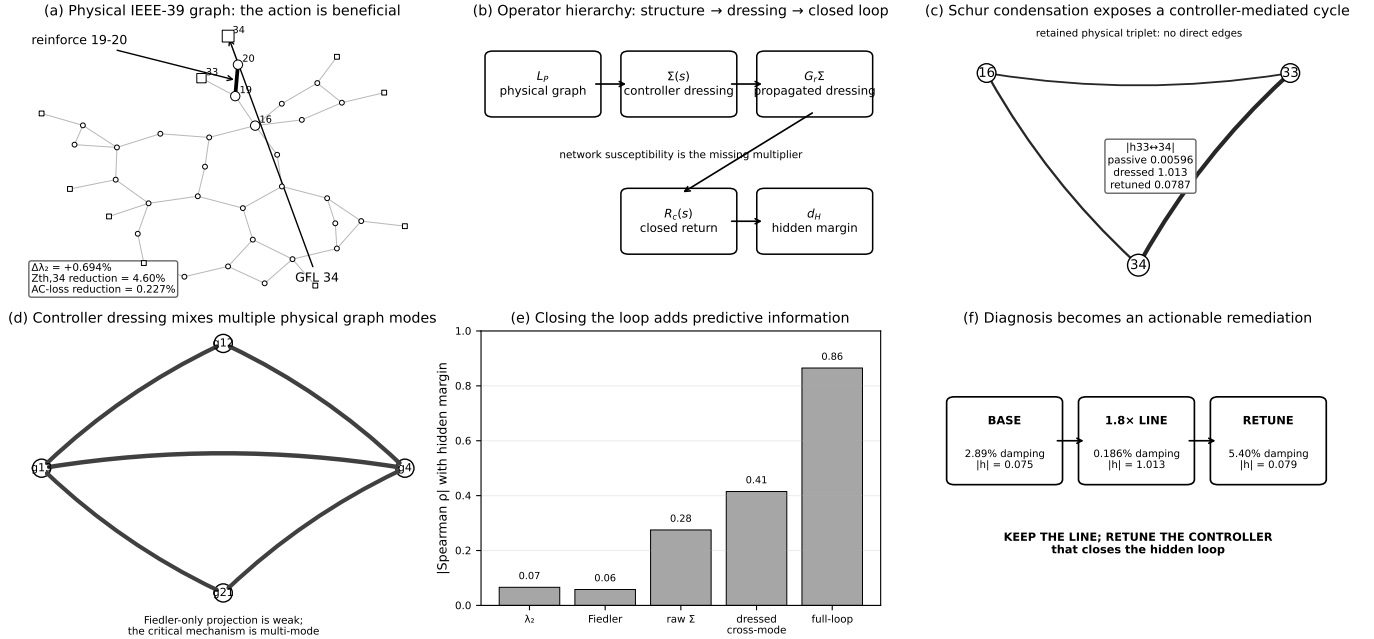


Fig. 1. Graphical summary of the proposed physical-to-dynamic hierarchy. (a) The IEEE 39-bus physical graph and the 19–20 star action are evaluated by AC quantities such as λ_2 , Z_{th} , and losses. (b) Controller self-energy is not used in isolation; network susceptibility propagates it through $G_T\sigma$ before the controller-side return closes through R_c . (c) Schur condensation exposes effective edges and a controller-mediated cycle absent from the retained physical triplet. (d) The same dressing mixes several physical graph modes rather than only the Fiedler mode. (e) Empirical information increases as the loop is closed. (f) The resulting diagnosis is actionable: retain the physically useful reinforcement and retune the controller that closes the hidden loop.

TABLE I
PRE-SPECIFIED HYPOTHESES, FALSIFICATION CRITERIA, AND EVIDENCE USED IN THE PROOF-OF-CONCEPT. EXACT ALGEBRAIC IDENTITIES ARE
SEPARATED FROM MODEL-DEPENDENT EMPIRICAL CLAIMS.

ID	Hypothesis	Falsification criterion	Evidence in this paper
H1	A physically beneficial action can reduce dynamic feedback margin.	No action with $\Delta\lambda_2 > 0$ and improved AC metrics yields $\Delta d_H < 0$.	19–20: $\Delta\lambda_2 = +0.694\%$, $-\Delta Z_{th} = 4.60\%$, loss reduction 0.227%, but $\Delta d_H = -91.3\%$.
H2	The reversal is controller-mediated rather than a passive Kron artifact.	Passive and controller-dressed reductions exhibit comparable cycle amplification, or retuning cannot restore margin.	Passive cycle 0.00596; dressed cycle 1.013; retuned cycle 0.0787 while the line remains reinforced.
H3	The mechanism is not unique to a single action magnitude or one line.	The line ranking changes strongly with action strength, or only 19–20 exhibits the effect.	Top-three continuous sweeps plus 46-line ranking; minimum rank correlation 0.996 across $1.4\text{--}2.0\times$.
H4	The diagnosis is not an artifact of the retained-port set.	Reinforcement ceases to reduce effective closure distance for reasonable port choices.	All five retained-port sets move closer to closure after reinforcement and farther after retuning.
H5	The reversal persists under operating-point/controller uncertainty.	Small load/tuning changes remove the damping degradation.	Load factor 0.85–1.15 maps and 300 parameter draws; among 223 base-stable draws, damping degrades in 100% and 31.4% cross instability.
H6	Closed-loop information outperforms physical/Fiedler-only screens.	Bootstrap intervals for the proposed feedback quantities overlap zero as strongly as physical baselines.	95% bootstrap CIs exclude zero for feedback-dressed cross-mode coupling and full-loop spectral radius, but not for λ_2 , Fiedler-only, or raw Σ mixing.
H7	Linear feedback screening transfers to a non-linear response metric across the complete action set.	Applying the same nonlinear forcing to all 46 actions destroys the linear ranking or yields correlations indistinguishable from zero.	46 actions $\times$ 80 trajectories/action under identical broadband OU forcing: $\rho(d_H, \Delta\text{RMS}) = -0.851$ and $\rho(\rho(R_c), \Delta\text{RMS}) = 0.994$; 4000-bootstrap CIs exclude zero for the feedback-dressed quantities.

topology describes dynamics that are not visible from the physical Laplacian alone.

Proposition 2 (Network/controller spectral equivalence): $K_H(s)$ and $R_c(s)$ have identical nonzero eigenvalues, including algebraic multiplicities.

Proof: Write $X = G_r A_{rc}$ and $Y = G_c A_{cr}$. Then $K_H = XY$ and $R_c = YX$. The nonzero spectra of XY and YX coincide. ■

This result justifies analyzing the interaction in either network coordinates (useful for GSP and localization) or controller coordinates (useful for low-dimensional loop screening).

B. Hidden effective edges and cycles

Partition a dynamic operator into retained ports b and hidden states h :

$$T(s) = \begin{bmatrix} T_{bb} & T_{bh} \\ T_{hb} & T_{hh} \end{bmatrix}. \quad (14)$$

Eliminating h gives

$$\boxed{T_{\text{eff}}(s) = T_{bb} - T_{bh}T_{hh}^{-1}T_{hb}. \quad (15)}$$

An off-diagonal element of T_{eff} can be nonzero even when the corresponding retained ports have no direct physical edge.

Such an element is called an *effective edge*; if it changes only after controller dressing it is controller-mediated rather than a purely passive Kron edge.

For a three-port normalized operator $T_{\text{eff}} = D(I - W)$ with $\text{diag}(W) = 0$, direct determinant expansion gives

$$\det(I - W) = 1 - W_{12}W_{21} - W_{13}W_{31} - W_{23}W_{32} - W_{12}W_{23}W_{31} - W_{13}W_{32}W_{21}. \quad (16)$$

Thus the interaction determinant is exactly decomposed into three directed 2-cycles and two oriented 3-cycles. For larger scalar-port graphs, the Leibniz expansion yields signed collections of vertex-disjoint cycles, while the log-determinant trace expansion organizes closed walks.

C. Rank-one cycle closure and topology updates

When $A_{rc} = bq^T$ and $A_{cr} = uc^T$, the return ratio has rank at most one. Its only possible nonzero eigenvalue is

$$h(s) = \underbrace{q^T G_c(s) u}_{G_c^{\text{loop}}(s)} \underbrace{c^T G_r(s) b}_{G_n^{\text{loop}}(s)}, \quad (17)$$

and

$$\det(sI - A) = \det T_r \det T_c [1 - h(s)]. \quad (18)$$

Hence an interaction-induced mode satisfies $h(s) = 1$ when s is not already a local pole.

At a frozen operating point, let a physical action update the network operator by rank one,

$$T_r(p) = T_r + pu_e v_e^T. \quad (19)$$

Sherman–Morrison gives

$$G_r(p) = G_r - \frac{pG_r u_e v_e^T G_r}{1 + pv_e^T G_r u_e}. \quad (20)$$

Substitution into (17) yields the exact finite-action map

$$h(p, s) = h(0, s) - \frac{pG_c^{\text{loop}}(s)[c^T G_r u_e][v_e^T G_r b]}{1 + pv_e^T G_r u_e}. \quad (21)$$

For a full AC action, (21) is the direct frozen-point contribution; the total derivative also contains the operating-point shift from the implicit-function theorem.

D. GSP view of controller-mediated mode mixing

Let U be the physical Laplacian basis. A raw controller-dressing map is

$$\widehat{\Sigma}(j\omega) = U^H \Sigma_{\text{bus}}(j\omega) U. \quad (22)$$

Its off-diagonal terms quantify controller-induced mixing between physical graph modes. The experiments below show that raw $\widehat{\Sigma}$ is not sufficient. The more informative object is the feedback-dressed projection

$$\widehat{K}_H(j\omega) = U^H K_{H,\text{bus}}(j\omega) U, \quad (23)$$

which includes both controller dressing and frequency-dependent network propagation. We use

$$g_{\max} = \max_{k \neq \ell} |[\widehat{K}_H]_{k\ell}| \quad (24)$$

as an interpretable cross-mode diagnostic rather than as a replacement for the exact hidden margin (11).

IV. IEEE 39-BUS NUMERICAL CASE AND FALSIFICATION PROTOCOL

A. Exact numerical model and reproducibility

The physical case is the standard MATPOWER IEEE 39-bus system. The AC power flow is re-solved after every topology action. A Newton–Raphson implementation from the published case data reproduces the stored MATPOWER solution with maximum voltage error 4.80×10^{-8} p.u., maximum angle error 4.95×10^{-7} degrees, and infinity-norm residual 1.43×10^{-13} p.u. This establishes that the physical-network metrics below are evaluated on the intended IEEE 39-bus operating point rather than on a separate approximate load-flow model.

The star numerical case is shown in Fig. 2. Bus 20 carries 680 MW and 103 Mvar in the MATPOWER case. Branch 19–20 has $r = 7 \times 10^{-4}$ p.u., $x = 0.0138$ p.u., and tap 1.06; bus 34 is connected to bus 20 through a branch with $r = 9 \times 10^{-4}$ p.u., $x = 0.0180$ p.u., and tap 1.009. In the dynamic proof-of-concept the synchronous unit at bus 34 is replaced by one stylized grid-following converter. The controller is intentionally transparent:

$$\dot{\phi} = \omega_{\text{pll}}, \quad (25)$$

$$\dot{\omega}_{\text{pll}} = -2\zeta_{\text{pll}}\omega_n\omega_{\text{pll}} - \omega_n^2 \sin(\phi - \theta_{34}), \quad (26)$$

$$\tau \dot{p}_c = -p_c + \text{sat}_{\bar{p}} [-K_e \sin(\phi - \theta_{34}) - K_\omega \omega_{\text{pll}} + u_d]. \quad (27)$$

The nominal parameters are $\omega_n = 17.9016$ rad/s, $\zeta_{\text{pll}} = 0.7874$, $K_e = 98.220$, $K_\omega = 1.65$, $\tau = 0.04956$ s, network damping coefficient $D_g = 2.8324$, and nonlinear limit $\bar{p} = 0.6$ p.u. The retuned case changes only K_ω from 1.65 to 2.20. These parameters are a stress-test surrogate, not a vendor-calibrated GFL model; critically, they are fixed before the 46-action atlas is generated. After the linear action atlas is frozen, every one of the 46 branch actions is also simulated under the *same* nonlinear stochastic protocol: 80 trajectories/action, 8 s horizon, 5 ms integration step, identical random seed, OU disturbance standard deviation 0.025 p.u., and no action-specific controller tuning. The response variables are median controller-action RMS, 95th-percentile controller peak, PLL-error RMS, and the probability of the same nonlinear event definition used in the earlier stress tests. This all-action run is used only as an internal cross-regime falsification test; it is not presented as independent model validation.

B. Numerical walk-through: safe $\rightarrow$ near closure $\rightarrow$ unstable $\rightarrow$ repaired

The central case is a continuous reinforcement of branch 19–20, parameterized by the admittance multiplier α . Table II gives the full numerical ledger. The sequence is deliberately explicit. At $\alpha = 1$ the controller-family mode is stable with 2.89% damping. At $\alpha = 1.8$, all three AC/network metrics improve, but the hidden feedback margin falls from 2.87×10^{-2} to approximately 2.05×10^{-3} and the controller-family damping falls to 0.186%. Near $\alpha = 1.925$ the pole crosses the imaginary axis. At $\alpha = 2$, corresponding to an identical parallel-admittance equivalent in this surrogate,

IEEE 39-bus numerical mechanism: a physical line action amplifies a hidden controller-mediated cycle

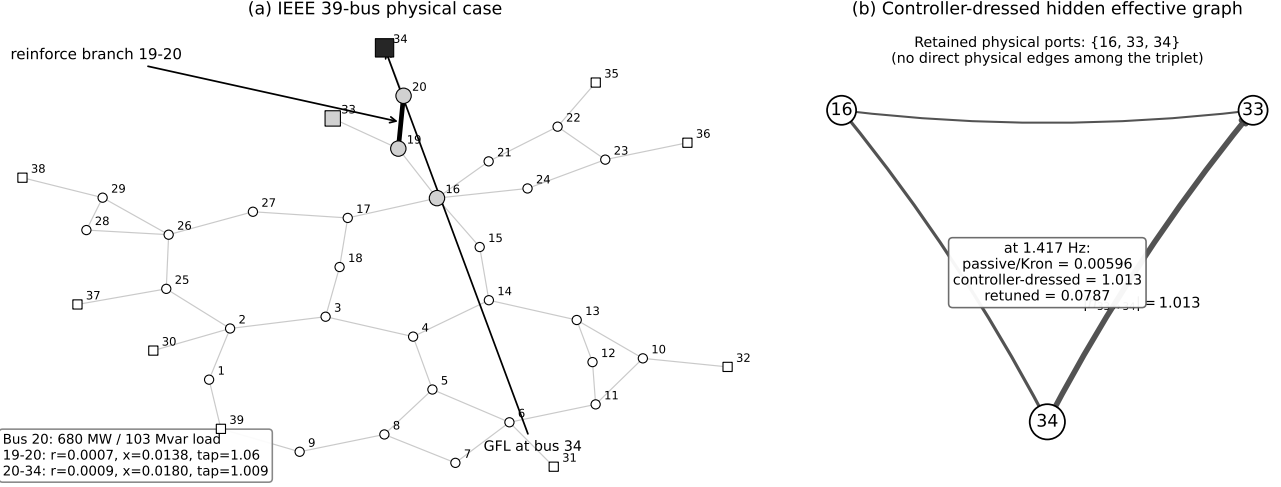


Fig. 2. IEEE 39-bus numerical mechanism. The physical action is an admittance increase on branch 19–20. Retaining only buses $\{16, 33, 34\}$ leaves no direct physical edges among the triplet, yet Schur condensation produces a controller-dressed effective graph. At 1.417 Hz the reinforced passive/Kron 33–34 cycle gain is only 0.00596, whereas controller dressing raises it to 1.013; controller retuning reduces it to 0.0787 without removing the line action.

$\Re\{\lambda_c\} = 9.67 \times 10^{-3} > 0$. Finally, restoring only controller damping at the same $\alpha = 1.8$ moves the critical controller pole to $-0.502 + j9.282$ rad/s and expands the hidden margin to 5.89×10^{-2} while leaving λ_2 , $Z_{th,34}$, and AC losses unchanged from the reinforced network.

Figure 3 places all six consequences on one numerical plate. The physical graph improves monotonically; the hidden margin and rank-one loop distance collapse; the controller-family pole moves through the imaginary axis; the hidden cycle peaks around the same frequency; removing controller feedback or retuning the controller suppresses the cycle; and the independent nonlinear stress test shows the same direction of risk.

C. Exact algebraic and numerical consistency checks

The proposed interpretation is not based on a fitted surrogate index. Several identities are checked independently at machine precision. For a frozen line update the network-operator perturbation is rank one. Direct inversion and Sherman–Morrison agree to relative error 4.84×10^{-16} ; the direct and updated return ratios agree to 1.18×10^{-16} ; and the characteristic determinant factorization in (12) agrees to 2.05×10^{-16} . Together with the AC power-flow residual above, these checks isolate modeling assumptions from algebraic/numerical error.

D. Exhaustive action atlas and blind nonlinear cross-check

A single counterexample would not justify a planning method. Therefore all 46 existing IEEE 39-bus branches are independently reinforced by $1.8\times$ and the AC operating point is re-solved for each action. The dynamic screen is then computed before nonlinear simulation. The three strongest predicted margin collapses are frozen as 19–20, 19–33, and 16–19; three electrically beneficial near-neutral controls are

frozen as 8–9, 29–38, and 4–5. No re-ranking is allowed after the nonlinear trajectories are generated.

The dangerous actions increase median nonlinear controller-action RMS by 52.9%, 32.5%, and 10.1%, whereas the benign controls remain at approximately zero change. The six-action rank correlation between predicted margin collapse and nonlinear RMS amplification is $\rho = 1.00$; event probability gives $\rho = 0.941$ ($p = 0.0051$). Halving the integration step preserves the dangerous/benign ordering. Figure 4 also shows why the result is not a Fiedler re-labeling: physical λ_2 and a Fiedler-only feedback projection are nearly uncorrelated with the hidden margin.

E. What Fiedler, GSP, self-energy, and loop closure each contribute

The exhaustive experiment produces an informative negative result. Across the 46 line actions,

$$\rho(\Delta\lambda_2, d_H) = 0.066, \quad (28)$$

$$\rho(\text{Fiedler-only projection}, d_H) = 0.058, \quad (29)$$

$$\rho(\text{raw } U^H \Sigma U, d_H) = 0.275, \quad (30)$$

$$\rho(g_{\max}(U^H K_H U), d_H) = -0.415, \quad (31)$$

$$\rho(\rho(R_c), d_H) = -0.865. \quad (32)$$

The feedback-dressed cross-mode correlation has $p = 0.0041$, and the full-loop correlation has $p = 9.38 \times 10^{-15}$. Thus the missing information is not another static Laplacian statistic and not controller self-energy in isolation. The controller dressing becomes predictive only after it is propagated by the network resolvent and closed through the feedback loop. This gives the hierarchy

$$L_P \rightarrow \Sigma(s) \rightarrow G_r(s)\Sigma(s) \rightarrow R_c(s) \rightarrow d_H. \quad (33)$$

TABLE II
NUMERICAL LEDGER FOR THE IEEE 39-BUS BRANCH 19–20 CASE. CYCLE GAIN IS EVALUATED AT THE CORRESPONDING CONTROLLER-FAMILY MODAL FREQUENCY.

Case	α	λ_2	$Z_{th,34}$	Loss (MW)	d_H	$\Re\{\lambda_c\}$	f_c (Hz)	ζ_c (%)	$ h_{33 \leftrightarrow 34} $
Base	1.000	5.3867	0.13114	43.641	2.865×10^{-2}	-0.24809	1.3665	2.888	0.134
Reinforced	1.800	5.4241	0.12511	43.542	2.047×10^{-3}	-0.01660	1.4169	0.186	1.013
Near crossing	1.925	5.4267	0.12462	43.534	6.50×10^{-4}	+0.00062	1.4225	-0.0069	1.450
Parallel-admittance equiv.	2.000	5.4281	0.12436	43.530	1.456×10^{-3}	+0.00967	1.4256	-0.108	1.747
Reinforced + retuned	1.800	5.4241	0.12511	43.542	5.886×10^{-2}	-0.50176	1.4773	5.398	0.082

IEEE 39-bus numerical walk-through: stronger physical network, weaker hidden feedback, then controller-only recovery

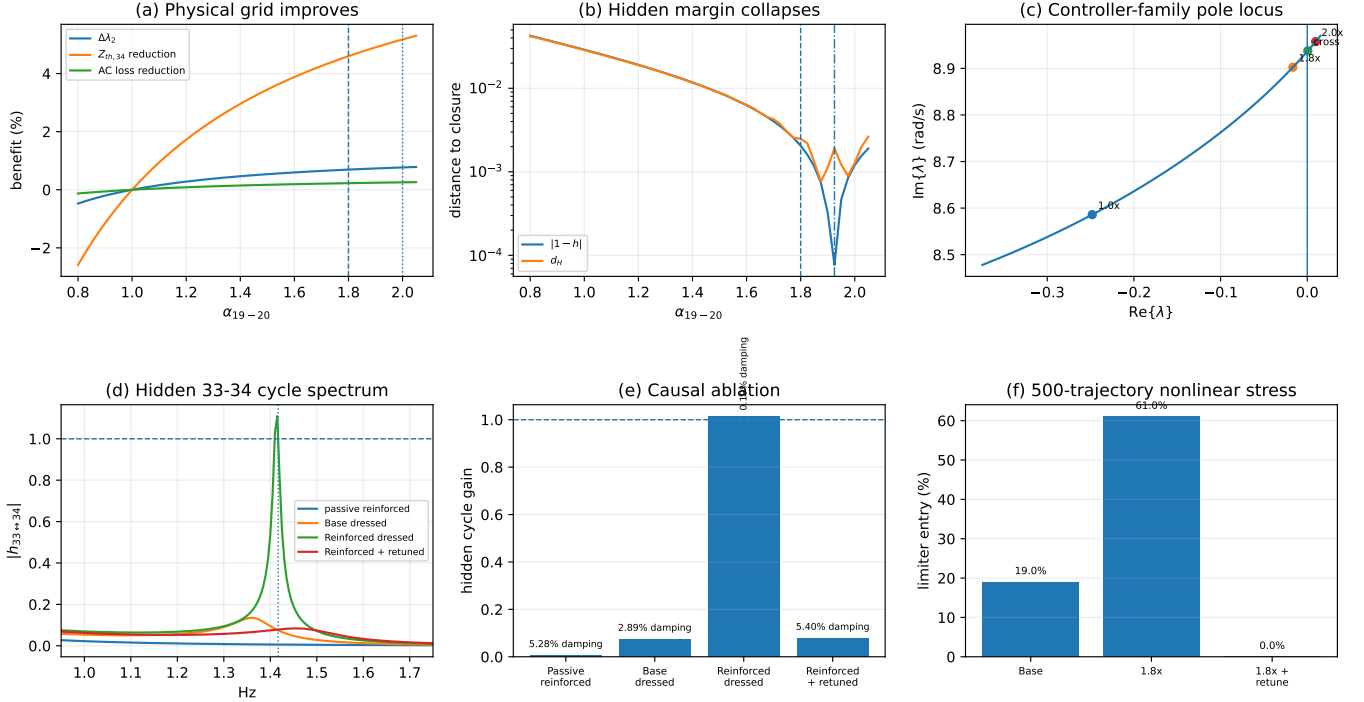


Fig. 3. Dense numerical walk-through of the branch 19–20 case. (a) Physical connectivity, Thevenin-impedance proxy, and AC losses all improve with reinforcement. (b) The feedback quantities $|1-h|$ and d_H collapse. (c) The controller-family pole crosses into the right-half plane. (d) The hidden 33–34 effective cycle is negligible in the passive reduction, becomes resonant after controller dressing, and is suppressed by retuning. (e) Causal ablation isolates the controller-mediated mechanism. (f) In a separate 500-trajectory resonance-matched nonlinear stress test, sustained current-limit entry rises from 19.0% to 61.0% after reinforcement and falls to 0/500 after retuning.

TABLE III
CONSISTENCY CHECKS USED TO SEPARATE ALGEBRAIC IDENTITY FROM MODELING ASSUMPTION

Check	Value
AC PF $\ F\ _\infty$	1.43×10^{-13}
Max MATPOWER $ V $ mismatch	4.80×10^{-8} p.u.
Max MATPOWER angle mismatch	4.95×10^{-7} deg
$\text{rank}(\Delta T_r)$	1
Woodbury relative error	4.84×10^{-16}
Return-ratio update error	1.18×10^{-16}
Characteristic-factor error	2.05×10^{-16}

TABLE IV
FROZEN PREDICTIONS VERSUS NONLINEAR CROSS-CHECK

Action	$\Delta\lambda_2$ (%)	Δd_H (%)	RMS amp. (%)	Event prob.
19–20	0.694	-91.3	+52.9	0.115
19–33	0.412	-62.0	+32.5	0.050
16–19	2.767	-23.9	+10.1	0.020
8–9	0.390	≈ 0	≈ 0	0.010
29–38	0.150	≈ 0	≈ 0	0.010
4–5	3.198	≈ 0	≈ 0	0.010

Spatially, the distinction is equally sharp. For 19–20 at 1.8 $\times$, the cosine overlap between Fiedler support and the critical hidden-mode support is 0.280. The critical hidden mode is concentrated at bus 34 (88.7%), bus 20 (9.0%), and bus 19

(2.0%), whereas the Fiedler vector distributes support across a broad cut structure. For 16–19 the overlap falls to 0.145. Fiedler analysis is therefore retained as the physical-structure baseline, while K_H and R_c describe the dynamic red zone and the closed-loop mechanism.

Why the result is not a single cherry-picked line: exhaustive ranking, blind nonlinear cross-check, and spatial/model hierarchy

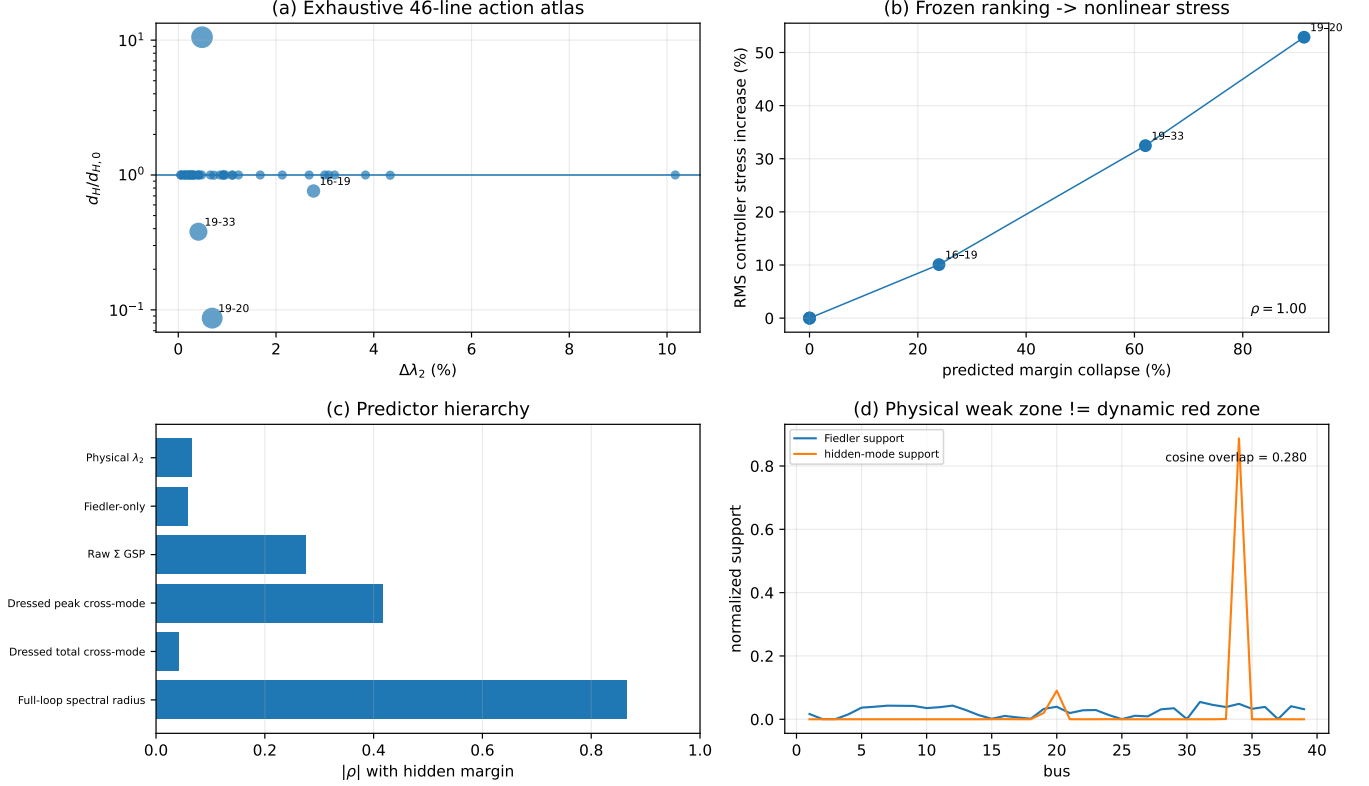


Fig. 4. Falsification-oriented screening plate. (a) Exhaustive 46-line atlas: electrically beneficial actions can have radically different feedback margins. (b) A six-action ranking frozen before nonlinear simulation predicts controller-stress amplification exactly in rank order. (c) Predictor hierarchy: physical λ_2 and Fiedler-only projections carry little information about d_H ; the full-loop spectral radius is strongest. (d) In the star case, the physical Fiedler weak-zone and critical hidden-mode support have cosine overlap only 0.280.

V. ROBUSTNESS, GENERALIZATION, AND STATISTICAL FALSIFICATION

The star case is useful only if the mechanism survives attempts to break it by changing controller tuning, loading, action identity and magnitude, reduced-port choice, stochastic spectrum, and uncertain parameters. We therefore treat robustness as a falsification program rather than as a collection of additional examples.

A. Deterministic robustness and generalization

Figure 5 consolidates four independent deterministic stress tests. First, panels (a)–(b) sweep branch-19–20 reinforcement against the damping-channel gain K_ω . The zero contour of $\Re\{\lambda_c\}$ forms a continuous stability boundary. At the pre-boundary reinforcement $\alpha = 1.8$, the same physical network ranges from unstable at $K_\omega \lesssim 1.62$ to strongly damped at larger gains; $K_\omega = 1.65$ gives $\zeta_c = 0.186\%$, while $K_\omega = 2.20$ gives 5.40% . The hidden margin is minimized near the pole crossing and then re-expands after the loop passes the boundary. Because d_H is an *unsigned distance to closure*, it must therefore be interpreted jointly with pole sign or an equivalent Nyquist winding when points on opposite sides of instability are compared.

Second, panels (c)–(d) sweep load factor from 0.85 to 1.15 and branch 19–20 from 1.0 to $2.0\times$. Every unreinforced load case is small-signal stable. At $\alpha = 1.8$, the controller pole ranges from $+0.0233$ at load factor 0.85 to -0.0528 at load factor 1.15. Across all 273 sampled load/action points, 95.2% simultaneously have $\Delta\lambda_2 > 0$ and a hidden margin below the same-load unreinforced baseline; the maximum AC power-flow residual is 3.28×10^{-10} p.u. Thus the severity varies with operating point, but the physical/dynamic disagreement is not confined to the nominal dispatch.

Third, panel (e) follows the three actions frozen as most dangerous by the $1.8\times$ screen—19–20, 19–33, and 16–19—continuously in action magnitude. They differ quantitatively but share the same qualitative separation between physical strengthening and hidden-margin degradation. Independently, Fig. 6(c) shows that the full 46-line hidden-margin ranking is almost invariant to the chosen action magnitude: the minimum pairwise Spearman correlation between the rankings at 1.4, 1.6, 1.8, and $2.0\times$ is 0.9957.

Fourth, panel (f) tests five retained-port sets: $\{16, 33, 34\}$, $\{19, 20, 34\}$, $\{16, 19, 20, 34\}$, $\{19, 20, 33, 34\}$, and $\{16, 19, 20, 33, 34\}$. At 1.417 Hz, every set moves closer to effective closure after reinforcement and farther from closure after retuning, while the passive reduction remains much

farther from closure than the controller-dressed reduction. Individual effective edges are port-dependent, as expected, but the diagnosis of a near-closing reduced feedback structure is not an artifact of one convenient triplet.

B. Parametric, statistical, and nonlinear falsification

Figure 6 collects uncertainty and nonlinear tests that are deliberately different from the deterministic maps. Controller parameters are perturbed independently around the nominal values ($\pm 5\%$ for ζ , K_e , K_ω , τ , and D_g ; $\pm 2.5\%$ for ω_n). Of 300 draws, 223 are stable before reinforcement. Conditional on those base-stable draws, the $1.8\times$ reinforcement reduces damping in all 223 [panel (a)], moves 83.4% closer to hidden closure, and drives 31.4% across the small-signal boundary [panel (b)]. The pole sign is retained as the primary stability classification because the unsigned distance d_H can re-expand after a draw has crossed the boundary.

Point correlations over only 46 actions are also assigned uncertainty rather than reported as deterministic facts. We bootstrap the action set and recompute Spearman correlations with d_H [panel (d)]. Physical λ_2 , Fiedler-only projection, and raw self-energy mixing all have 95% intervals that cross zero. The feedback-dressed cross-mode diagnostic has median $\rho = -0.418$ with interval $[-0.668, -0.094]$, while the full-loop spectral radius has median $\rho = -0.874$ with interval $[-0.992, -0.666]$. Panel (c) separately reports the action-ranking stability noted above.

Finally, the original 500-trajectory resonance-matched nonlinear test is supplemented by a multi-spectrum experiment at disturbance centers 0.8, 1.417, and 2.2 Hz, with 300 trajectories per case and identical random seeds. Reinforcement increases controller-action RMS in all three bands and retuning suppresses it, with bootstrap confidence intervals [panel (e)]. The independent higher-resolution matched-frequency experiment gives the stronger limiter-entry sequence $19.0\% \rightarrow 61.0\% \rightarrow 0/500$ [panel (f)]. The off-resonance and near-critical tests therefore support a frequency-selective amplification mechanism without relying on one stochastic realization or one event threshold.

C. All-action cross-regime nonlinear validation

The six-action frozen test in Table IV deliberately separates three predicted dangerous actions from three benign controls. We additionally remove that selection step and simulate *all* 46 branch actions under the common nonlinear OU protocol described above. Figure 7 reports this complete cross-regime test. The hidden margin is strongly anti-correlated with nonlinear controller-action RMS change, $\rho = -0.851$ ($p = 6.42 \times 10^{-14}$), while the full-loop spectral radius nearly preserves the nonlinear ranking, $\rho = 0.994$ ($p = 7.29 \times 10^{-44}$). In contrast, physical λ_2 has $\rho = -0.048$ ($p = 0.752$), a Fiedler-only feedback projection has $\rho = -0.149$ ($p = 0.322$), and raw Σ GSP mixing has $\rho = -0.181$ ($p = 0.230$). Feedback-dressed cross-mode coupling is intermediate but significant, $\rho = 0.367$ ($p = 0.012$).

The ordering is not driven by a few selected points. Four-thousand-bootstrap 95% intervals for correlation with

TABLE V
ROBUSTNESS SUMMARY; ALL VALUES ARE MODEL-DEPENDENT
PROOF-OF-CONCEPT EVIDENCE.

Test	Quantitative result
Operating point	13 load factors $\times$ 21 line multipliers; 95.2% have $\Delta\lambda_2 > 0$ and d_H below same-load base; PF residual $< 3.3 \times 10^{-10}$ p.u.
Controller tuning	21 K_ω values $\times$ 22 line multipliers; at $\alpha = 1.8$, ζ_c spans -2.70% to $+6.79\%$.
Port selection	Five retained sets; all move closer to effective closure after reinforcement and farther after retuning.
Parameter uncertainty	300 draws; 223 base-stable; 100% lose damping, 83.4% move closer in d_H , 31.4% cross instability.
Action magnitude	46 lines at 1.4, 1.6, 1.8, $2.0\times$; minimum pairwise ranking correlation 0.9957.
Predictor bootstrap	2500 resamples; λ_2 , Fiedler-only, raw Σ : CIs cross zero; feedback-dressed cross-mode and full loop: CIs exclude zero.
All-action nonlinear transfer	46 lines $\times$ 80 trajectories/action; $\rho(d_H, \Delta\text{RMS}) = -0.851$, $\rho(\rho(R_c), \Delta\text{RMS}) = 0.994$; full-loop 4000-bootstrap CI $[0.977, 0.998]$.

nonlinear RMS are $[-0.337, 0.237]$ for λ_2 , $[-0.503, 0.210]$ for Fiedler-only, $[-0.495, 0.159]$ for raw Σ , $[0.050, 0.622]$ for feedback-dressed cross-mode coupling, $[0.977, 0.998]$ for the full-loop spectral radius, and $[-0.981, -0.650]$ for d_H . The same three actions frozen as most dangerous in the linear screen—19–20, 19–33, and 16–19—are the three largest nonlinear RMS amplifiers in the complete 46-action run, at $+62.5\%$, $+34.6\%$, and $+10.3\%$, respectively. Nonlinear event probability is also ordered by d_H ($\rho = -0.489$, $p = 5.67 \times 10^{-4}$). This result is stronger than the six-action cross-check because no line is excluded after the linear prediction is made; nevertheless, it remains an internal surrogate-model validation and therefore does not replace the frozen ANDES/EMT test.

Deterministic robustness across tuning, loading, actions, and retained ports

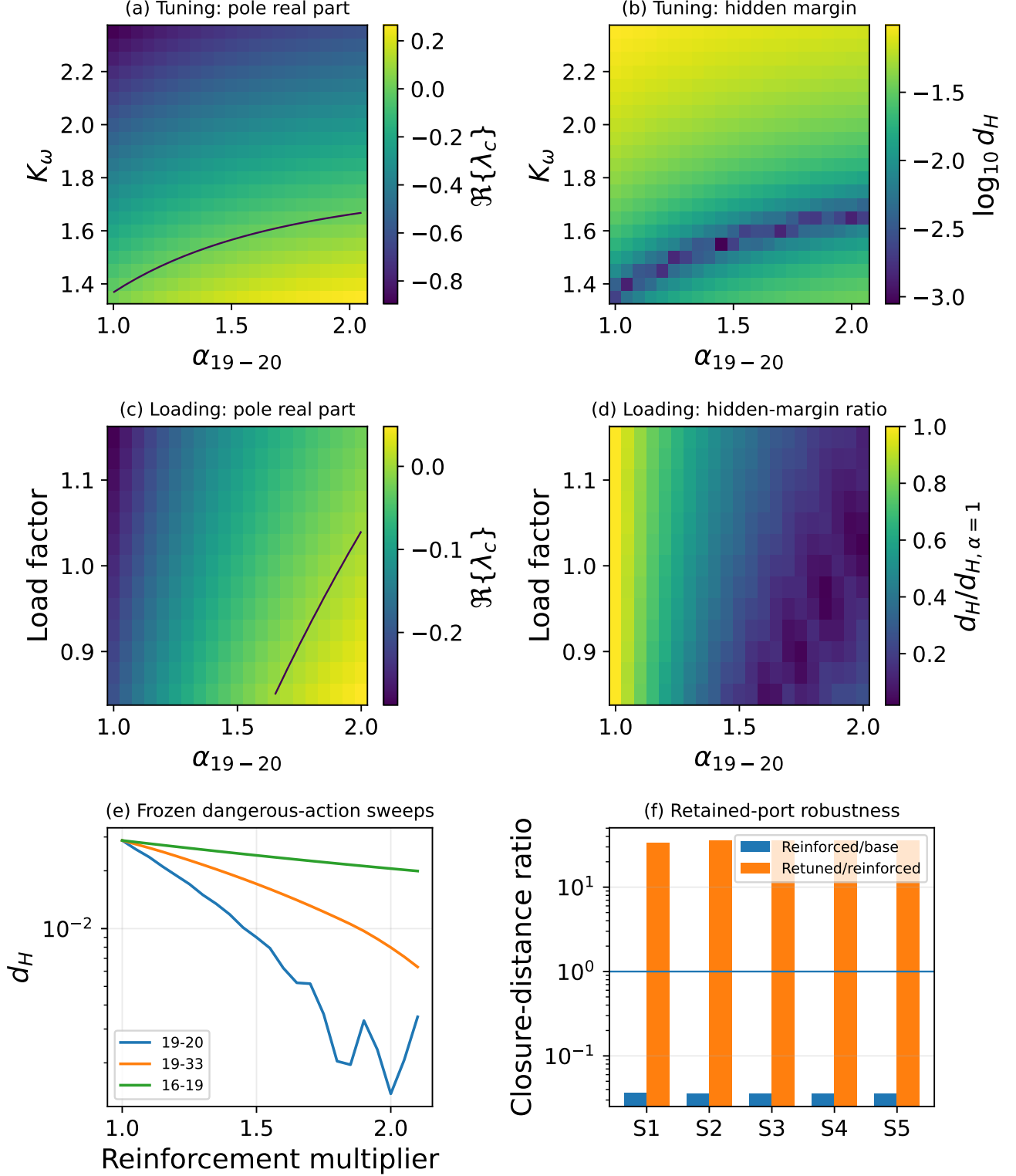


Fig. 5. Deterministic falsification tests for the controller-mediated reinforcement reversal. (a)–(b) Controller tuning K_ω versus branch-19–20 reinforcement: controller-family pole real part and hidden feedback margin d_H . (c)–(d) Loading versus reinforcement: pole real part and hidden-margin ratio relative to the same-load base. (e) Continuous hidden-margin sweeps of the three actions frozen as most dangerous by the $1.8\times$ screen. (f) Retained-port robustness: every tested reduction moves toward closure after reinforcement (blue ratio < 1) and away after retuning (orange ratio > 1). Together the panels test tuning, operating point, action identity, action magnitude, and port choice without changing the hypothesized mechanism.

Statistical and nonlinear falsification tests

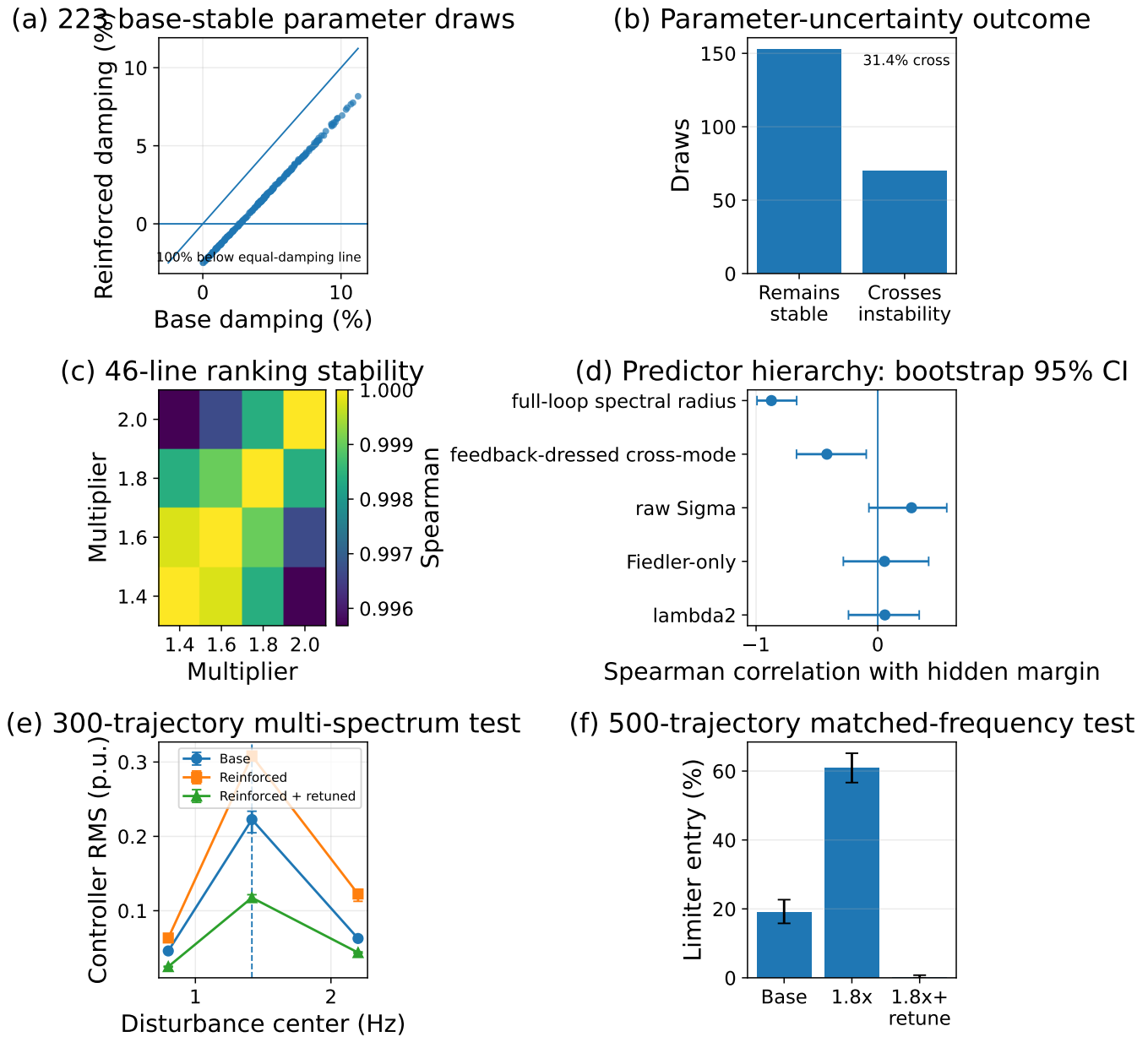


Fig. 6. Statistical and nonlinear falsification tests. (a) Reinforced versus base damping for the 223 parameter draws that are base-stable; all lie below the equal-damping line. (b) Of these draws, 31.4% cross instability after the same physical reinforcement. (c) Pairwise Spearman correlations of the 46-line hidden-margin ranking at four reinforcement magnitudes; all are at least 0.9957. (d) Bootstrap 95% intervals for the predictor hierarchy: only feedback-dressed cross-mode coupling and the full-loop spectral radius robustly exclude zero. (e) Controller-action RMS under 300-trajectory off-resonance and near-critical stochastic tests, with bootstrap confidence intervals. (f) Independent 500-trajectory matched-frequency limiter-entry experiment with uncertainty bars. The panels combine parametric, ranking, statistical, spectral, and nonlinear stress tests.

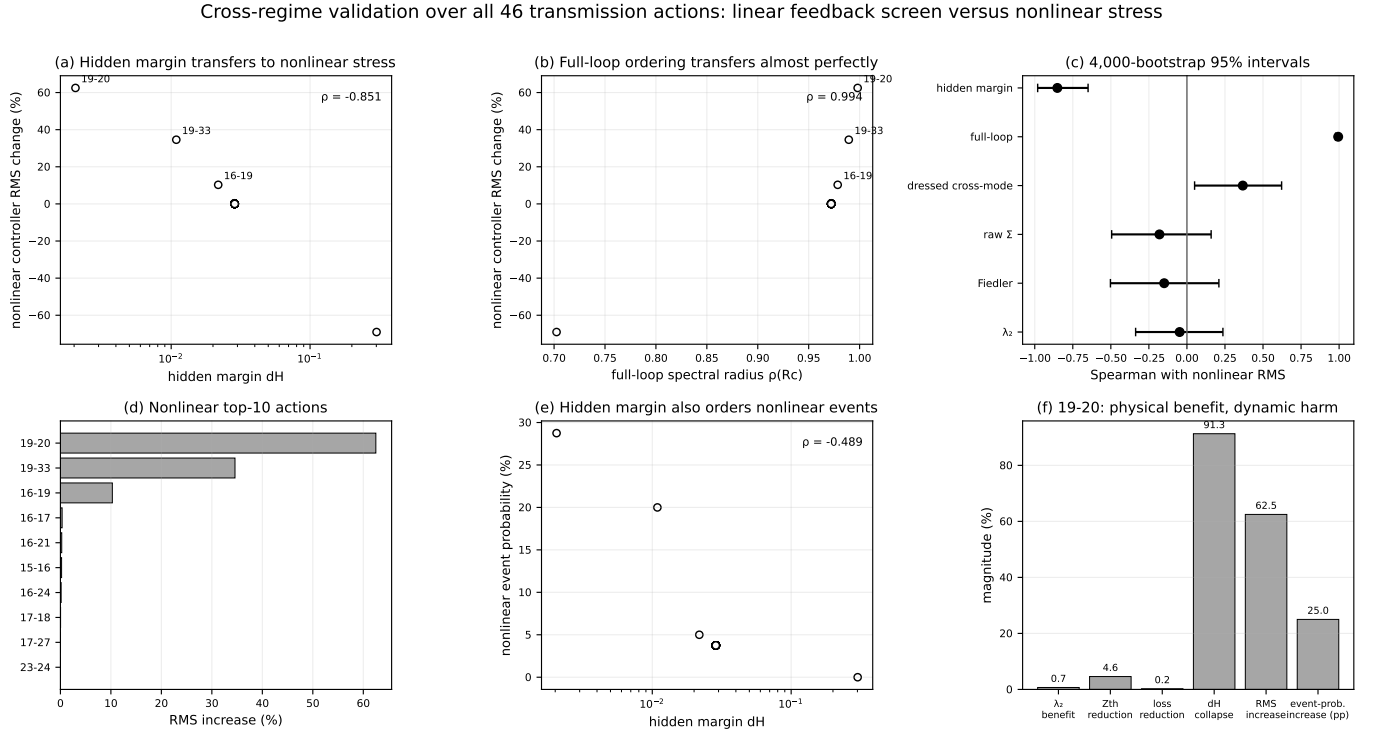


Fig. 7. Complete 46-action cross-regime nonlinear validation. (a) Hidden feedback margin versus nonlinear controller-action RMS change under an identical broadband OU disturbance for every reinforced branch. (b) Full-loop spectral radius versus the same nonlinear response; the ranking transfers with $\rho = 0.994$. (c) Four-thousand-bootstrap 95% intervals for correlations with nonlinear RMS: physical λ_2 , Fiedler-only projection, and raw Σ are not statistically separated from zero, whereas feedback-dressed cross-mode coupling, full-loop spectrum, and hidden margin are. (d) The top ten nonlinear stress amplifiers; the three actions frozen as dangerous by the linear screen remain the top three. (e) Hidden margin also orders nonlinear event probability. (f) The 19–20 star action summarized on a common scale: modest physical benefits accompany a 91.3% hidden-margin collapse, 62.5% RMS increase, and 25 percentage-point increase in event probability in this common broadband test.

VI. PLANNING AND INDUSTRIAL INTERPRETATION

The framework produces a decision structure that differs from a conventional weak-bus ranking. A candidate action can improve the physical network and still reduce the hidden margin. We refer to this situation as a *controller-mediated reinforcement reversal*. The operator should not necessarily reject the physical action: the causal ablation suggests a more useful workflow,

keep reinforcement + retune limiting controller. (34)

This interpretation supports several industrial uses.

- 1) **Transmission screening:** use AC metrics to remove physically unattractive projects, then use d_H and hidden-cycle diagnostics to identify actions requiring detailed dynamic studies.
- 2) **Commissioning:** determine whether a topology change requires PLL/current-loop retuning before energization.
- 3) **IBR/BESS siting:** compare candidate locations by their effect on both physical strength and hidden-cycle closure rather than only SCR or Fiedler participation.
- 4) **Dynamic red zones:** project critical hidden modes and effective cycles back onto buses, lines, and devices to identify the physical support of the margin-setting feedback sector.
- 5) **Model reduction:** reduce large regions to port operators while preserving the feedback-dressed cycles that control stability rather than preserving only static terminal behavior.

In industrial facilities, the same logic can be applied to a PCC feeding active front ends, variable-frequency drives, UPS systems, filters, PV inverters, and BESS. The physical one-line diagram may be radial, yet controller/network condensation can create an effective loop among devices. The relevant engineering question becomes not only “how strong is the PCC?” but “which controller-mediated loop is closest to closure at the frequencies excited by the installed equipment?”

VII. RELATION TO PRIOR WORK

The proposed framework is complementary to impedance-based and interaction-admittance stability analysis, not a replacement. Sun’s impedance criterion [2] and subsequent multivariable approaches establish the importance of converter/grid return ratios. Lu *et al.* [3] provide a physical admittance interpretation of multi-inverter interaction, while PLL/grid coupling and controller-side remedies are demonstrated in [4]. Recent holistic modeling retains full-order control and power-network dynamics in large inverter-intensive systems [10]; frequency-dependent system-strength work explicitly measures dynamic strength through impedance rather than a single static SCR [11]; and whole-system admittance identification targets practical closed-loop oscillation/root-cause analysis [12]. The present work differs by making the *planning action* explicit and by tracking how that action changes a feedback-dressed effective topology, its hidden cycles, and its physical support.

Similarly, Kron reduction [5] explains why passive node elimination produces effective edges. The controller-dressed

effective edges used here include that passive contribution but are separated experimentally from it. GSP [7] supplies the modal coordinates used to compare Fiedler structure with dynamic feedback mixing; the key empirical result is that raw graph modes or raw controller self-energy are not sufficient without the network resolvent. Graph-theoretic stability analysis has recently advanced to exact criteria for a class of lossless inverter grids [13]; our retained-device model and question are different, but that work reinforces the importance of separating graph structure, operating point, and device dynamics. Braess-type topology reversals are also known in oscillator networks [14]. The present claim is therefore not that adding a line can ever be harmful, but that a physical planning action can be mapped to a *controller-mediated effective cycle* whose closure explains and predicts the reversal. Finally, nonlinear synchronization results in oscillator networks [15] motivate graph-based reasoning for power-system dynamics, but the present object is a frequency-dependent closed-loop operator rather than a fixed physical coupling graph.

VIII. LIMITATIONS AND FALSIFIABLE NEXT TESTS

The results are intentionally framed as a proof-of-concept despite the extensive internal falsification tests. The dynamic model is not a validated production GFL model; the reported stochastic event probabilities are not outage probabilities; the controller parameters were chosen to expose the mechanism transparently; and all line-action, loading, uncertainty, and nonlinear checks—including the complete 46-action nonlinear run—share the same surrogate model class. Consequently, robustness *within* this model does not substitute for cross-model validation. The decisive next test is independent validation with detailed DAE models.

The ranking has therefore been frozen before such validation: 19–20, 19–33, and 16–19 are the predicted dangerous actions. In ANDES, the same ranking should be tested without re-selection using REGCP1/PLL dynamics, full power-flow/DAE relinearization, and time-domain disturbances. The falsifiable outcomes are: (i) whether the frozen ranking predicts damping degradation, (ii) whether d_H predicts critical-clearing-time or current/PLL excursion, and (iii) whether controller retuning restores transient margin while the physical reinforcement remains. EMT validation should follow only for the strongest cases.

A second extension is to evaluate $K_H(j\omega, x(t))$ along nonlinear trajectories. If hidden-cycle closure consistently precedes current limiting or loss of synchronization, the framework could become a transient early-warning variable. That extension is not claimed in the present paper.

IX. CONCLUSION

This paper argues that inverter-rich grids should be viewed through two complementary topologies: the physical topology that carries power and a frequency-dependent feedback-dressed topology that carries dynamic return. The controller self-energy $\Sigma(s)$ is necessary but not sufficient; the stability-relevant object is obtained after propagation through the network resolvent and closure of the network–controller loop.

From diagnosis to controller-aware transmission planning

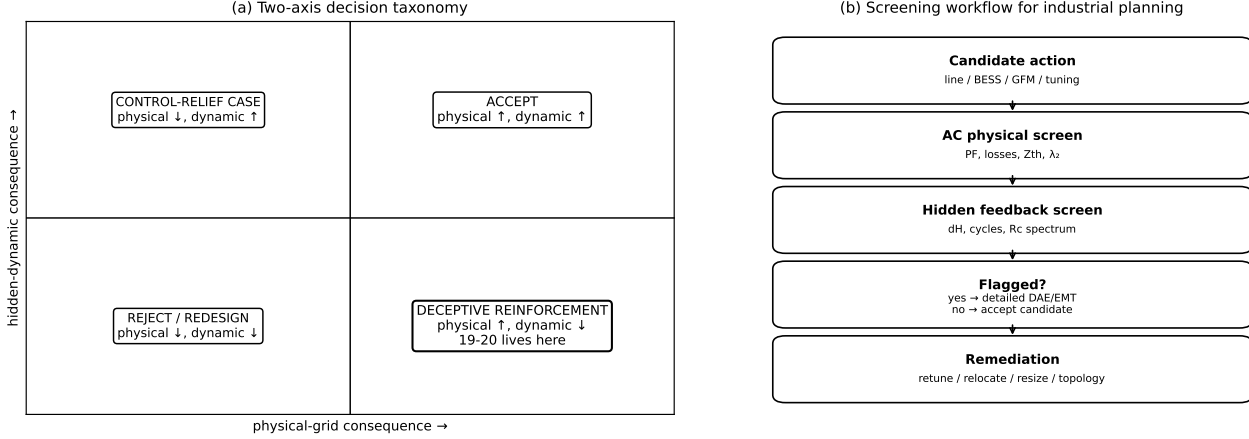


Fig. 8. Controller-aware planning interpretation. (a) Candidate actions should be classified on two axes: their physical-grid consequence and their hidden-dynamic consequence. The 19–20 star action is a deceptive reinforcement: physically beneficial but dynamically harmful under the nominal controller. (b) Proposed screening workflow: conventional AC metrics remove physically unattractive projects, hidden feedback quantities flag candidates requiring detailed DAE/EMT study, and the diagnosed mechanism determines whether the appropriate remediation is retuning, relocation, resizing, or topology redesign.

Exact determinant identities connect that object to interaction-induced modes, while Schur condensation exposes hidden effective edges and cycles.

In the IEEE 39-bus proof-of-concept, an electrically beneficial transmission reinforcement improves algebraic connectivity, network impedance, and AC losses while collapsing a controller-feedback margin and nearly eliminating damping. The critical effective cycle is negligible in the passive reduction, becomes near-unity after controller dressing, and is suppressed by controller retuning without removing the physical reinforcement. An exhaustive action screen is further tested on all 46 reinforced branches under identical nonlinear forcing: hidden margin and full-loop spectrum transfer strongly to nonlinear controller stress ($\rho = -0.851$ and 0.994 , respectively), whereas physical λ_2 , Fiedler projection, and raw self-energy mixing are substantially less informative and have bootstrap intervals that include zero. The reversal also survives loading, tuning, retained-port, action-strength, nonlinear-spectrum, and parameter-uncertainty tests; bootstrap intervals support the hierarchy from physical structure to feedback-dressed propagation to full-loop closure. These results motivate a controller-aware planning workflow in which physical strength is screened first and hidden dynamic topology is then used to identify deceptive reinforcements, critical devices, and targeted control remedies.

APPENDIX A COORDINATE INVARIANCE OF HIDDEN-STATE ELIMINATION

Let the hidden coordinates be transformed by $x'_h = Sx_h$ with nonsingular S . The transformed blocks satisfy $T'_{bh} = T_{bh}S^{-1}$, $T'_{hb} = ST_{hb}$, and $T'_{hh} = ST_{hh}S^{-1}$. Therefore

$$T'_{bh}(T'_{hh})^{-1}T'_{hb} = T_{bh}T_{hh}^{-1}T_{hb}. \quad (35)$$

Hence the retained effective operator (15) is invariant to arbitrary invertible coordinate changes internal to the eliminated

subsystem. Hidden effective edges defined at the retained ports are therefore not artifacts of the controller-state basis.

APPENDIX B SUBGRAPH COMPOSITION VIEW

Let $T_0(s) = \text{diag}(T_1(s), \dots, T_m(s))$ contain local subgraph operators and let $C(s)$ encode port couplings. Then

$$T = T_0 - C = T_0(I - G_0C), \quad G_0 = T_0^{-1}, \quad (36)$$

so that

$$T^{-1} = (I - G_0C)^{-1}G_0. \quad (37)$$

If $\rho(G_0C) < 1$,

$$T^{-1} = G_0 + G_0CG_0 + G_0CG_0CG_0 + \dots \quad (38)$$

The terms are ordered subgraph walks. Closed products contribute to the interaction determinant $\det(I - G_0C)$, providing a direct algebraic bridge between subgraph composition and feedback cycles.

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